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United States Patent	12407321
Kind Code	B2
Date of Patent	September 02, 2025
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Quartz-crystal vibrating piece, crystal unit, crystal controlled oscillator, and intermediate wafer for quartz-crystal vibrating piece

Abstract

A quartz-crystal vibrating piece includes an element unit, excitation electrodes, and an extraction electrode. The element unit includes a vibrator, a first thick portion, and second thick portions. The vibrator has a quadrilateral shape in plan view, vibrates in a thickness-shear mode, and has a thickness t_1 . The first thick portion has a rectangular shape in plan view, has a short side connected to one side of the vibrator, extends in a first direction, and has a thickness t_2 thicker than the thickness t_1 . The second thick portions are connected to both side surfaces of an end portion of the first thick portion in an opposite side of the vibrator. Each of the second thick portions extends along a second direction intersecting with the first direction and having a thickness t_3 thicker than the thickness t_1 and same as or different from the thickness t_2 .

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Appl. No.:	18/590917
Filed:	February 28, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240305264 A1	Sep. 12, 2024

Foreign Application Priority Data

JP	2023-036463	Mar. 09, 2023
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Publication Classification

Int. Cl.: **H03H9/02** (20060101); **H03B5/32** (20060101); **H03H9/05** (20060101); **H03H9/10** (20060101); **H03H9/19** (20060101)

U.S. Cl.:

CPC **H03H9/02133** (20130101); **H03B5/32** (20130101); **H03H9/02157** (20130101); **H03H9/0557** (20130101); **H03H9/1021** (20130101); **H03H9/19** (20130101);

Field of Classification Search

CPC: H03H (9/02133); H03H (9/02157); H03H (9/0557); H03H (9/1021); H03H (9/19); H03B (5/32)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is based on and claims priority under 35 U.S.C. § 119 to Japanese Patent Application No. 2023-036463, filed on Mar. 9, 2023, the entire content of which is incorporated herein by reference.

TECHNICAL FIELD

(2) This disclosure relates to a quartz-crystal vibrating piece appropriate for achieving a high frequency quartz crystal device, a crystal unit and a crystal controlled oscillator using the quartz-crystal vibrating piece, and an intermediate wafer for the quartz-crystal vibrating piece.

DESCRIPTION OF THE RELATED ART

(3) In order to cope with increase of the volume of communication in information communication, it is required to further increase the communication speed. Therefore, higher frequencies are desired for piezoelectric devices that are reference signal sources of the information communication.

(4) To attempt the higher frequency in a crystal unit and a crystal controlled oscillator that use a piezoelectric vibrating piece that vibrates in a thickness-shear mode, a thickness of a vibrator needs to be more and more thinned. However, when the thickness of the vibrator is thinned, an influence

of, for example, a stress on the vibrator from a supporting portion that supports the piezoelectric vibrating piece is concerned.

(5) To avoid this, for example, Japanese Unexamined Patent Application Publication No. 2001-144578 discloses a piezoelectric resonator with a structure designed for a high frequency piezoelectric device, for example, a frequency of 600 MHz. In the piezoelectric resonator, a vibrator formed of a thin piece and a supporting portion, which is provided along one side of an outer periphery of the vibrator and thicker than a plate thickness of the vibrator, are integrally formed (claim 1, FIG. 1, paragraph 2, and the like of Japanese Unexamined Patent Application Publication No. 2001-144578). According to the technique of Japanese Unexamined Patent Application Publication No. 2001-144578, since the thickness of the supporting portion can be configured to be any thickness regardless of an oscillation frequency, the thickness of the supporting portion can counteract the stress of deformation of a container, and the fluctuation of oscillation frequency can be avoided (paragraph 33 of Japanese Unexamined Patent Application Publication No. 2001-144578).

(6) However, through the actual development of a high frequency piezoelectric device, for example, exceeding 400 MHz, as a high frequency piezoelectric device, the study by the inventor of this application has revealed that the further innovation is necessary.

(7) A need thus exists for a quartz-crystal vibrating piece, a crystal unit, a crystal controlled oscillator, and an intermediate wafer for the quartz-crystal vibrating piece which are not susceptible to the drawback mentioned above.

SUMMARY

(8) According to a first aspect of this disclosure, there is provided a quartz-crystal vibrating piece that includes an element unit, excitation electrodes, and an extraction electrode. The element unit includes a vibrator, a first thick portion, and second thick portions. The vibrator has a quadrilateral shape in plan view, vibrates in a thickness-shear mode, and has a thickness t_1 . The first thick portion has a rectangular shape in plan view, has a short side connected to one side of the vibrator, extends in a first direction, and has a thickness t_2 thicker than the thickness t_1 . The second thick portions are connected to both side surfaces of an end portion of the first thick portion in an opposite side of the vibrator. Each of the second thick portions extends along a second direction intersecting with the first direction and has a thickness t_3 thicker than the thickness t_1 and same as or different from the thickness t_2 . The second thick portions are used for connecting the quartz-crystal vibrating piece to an external member. The excitation electrodes are disposed on both front and back sides of the vibrator. The extraction electrode is provided from the excitation electrode to the second thick portion via the first thick portion.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The foregoing and additional features and characteristics of this disclosure will become more apparent from the following detailed description considered with reference to the accompanying drawings, wherein:

(2) FIG. 1A to FIG. 1C are drawings for describing a quartz-crystal vibrating piece **10** of an embodiment of a first aspect;

(3) FIG. 2 illustrates an analysis result of a finite element method for describing a degree of an influence of a stress on the quartz-crystal vibrating piece **10** of the embodiment of the first aspect;

(4) FIG. 3 is a drawing for describing a crystal unit **30** of an embodiment of a second aspect;

(5) FIG. 4A is a drawing for describing characteristics of an oscillation frequency change over time of a crystal unit experimentally produced using the quartz-crystal vibrating piece **10** of the embodiment of the first aspect, and FIG. 4B is a drawing for describing characteristics of a drive

current change over time of the crystal unit;

(6) FIG. 5 is a drawing for describing a crystal controlled oscillator **40** of an embodiment of a third aspect;

(7) FIG. 6 is a drawing for describing an intermediate wafer **50** for a quartz-crystal vibrating piece of an embodiment of a fourth aspect; and

(8) FIG. 7A to FIG. 7E are drawings for describing Modification 1 to Modification 5 of the quartz-crystal vibrating piece of the first aspect.

DETAILED DESCRIPTION

(9) The following describes embodiments of respective aspects of this application with reference to the drawings. Each drawing used in the description is merely illustrated schematically for understanding this disclosure. In each drawing used in the description, the same reference numeral is attached to the similar component, and its explanation is omitted in some cases. Shapes, materials, and the like described in the following explanations are merely preferable examples within the scope of this disclosure. Therefore, this disclosure is not limited to only the following embodiments.

1. Embodiment of Aspect of Quartz-Crystal Vibrating Piece

(10) With reference to FIG. 1A to FIG. 1C, a quartz-crystal vibrating piece **10** of an embodiment of a first aspect will be described. Here, FIG. 1A is a perspective view of the quartz-crystal vibrating piece **10**, FIG. 1B is a side view of the quartz-crystal vibrating piece **10**, and FIG. 1C is a top view of the quartz-crystal vibrating piece **10**.

(11) The quartz-crystal vibrating piece **10** of the embodiment includes an element unit **11** including a vibrator **11a**, a first thick portion **11b**, and second thick portions **11c**, excitation electrodes **13**, and extraction electrodes **15**. The vibrator **11a**, the first thick portion **11b**, and the second thick portions **11c** are integrally formed, and for example, formed from an AT-cut crystal wafer by a photolithography technique. The following describes specific structures of respective parts.

(12) The vibrator **11a** has a quadrilateral shape in plan view, vibrates in a thickness-shear mode, and has a thickness $t1$. When the element unit **11** itself is configured of an AT-cut crystal element, the vibrator **11a** is also configured of the AT-cut crystal element.

(13) While the planar shape of the vibrator **11a** may be a square shape or a rectangular shape, a square-shaped one having one side with a length of α is used in this embodiment. The dimension α is determined corresponding to the design of the vibrator **11a**. The thickness $t1$ of the vibrator **11a** is a thickness corresponding to an oscillation frequency required for the quartz-crystal vibrating piece **10**. That is, when the required oscillation frequency is, for example, 460 MHz, the thickness $t1$ is approximately 3.5 μm , and it is considerably thin. However, this thickness is indicated with a value without considering a mass load of the excitation electrode **13**. When the thickness $t1$ of the vibrator **11a** is considerably thin, the excessively large dimension α causes a deficiency, such as a deflection of a vibrator **11a** itself. Therefore, the upper limit of the dimension α is preferred to be determined in consideration of this respect. When the required oscillation frequency is, for example, 460 MHz, for example, the upper limit of the dimension α of the vibrator **11a** is preferably about 0.5 mm, more preferably about 0.4 mm, although not limited thereto.

(14) The first thick portion **11b** has a rectangular shape in plan view, has a short side connected to one side of the vibrator **11a**, extends along a first direction **D1**, and has a thickness $t2$ thicker than the thickness $t1$ of the vibrator **11a**.

(15) Here, a dimension β along the first direction **D1** of a portion of the first thick portion **11b** excluding a portion in contact with the second thick portions **11c** is configured to be a dimension meeting $\beta > \alpha$. In the case of $\beta > \alpha$, compared with the other case, since the vibrator **11a** can be spaced from the second thick portions **11c** by at least a distance α , the mutual interference between the second thick portions **11c** secured to an external member and the vibrator **11a** is easily reduced, and it is preferred. The larger dimension β is preferred because the vibrator **11a** can be more spaced from the second thick portions **11c**, but the upper limit of the dimension β is determined in

consideration of the whole dimension in the first direction required for the quartz-crystal vibrating piece.

(16) In this embodiment, one surface of the vibrator **11a** and one surface of the first thick portion **11b** constitute a continuous flat surface. With this structure, for the formation of the vibrator **11a**, it is only necessary to etch a planned region for forming the vibrator **11a** of a crystal wafer (not illustrated) from one surface of the crystal wafer until a remaining thickness becomes t_1 , and it is preferred.

(17) In this embodiment, the first thick portion **11b** includes an inclined portion **11ba** having a thickness gradually thinned toward the vibrator **11a** in the vibrator **11a** side. When the inclined portion **11ba** is provided, the mutual interference between a portion excluding the inclined portion **11ba** of the first thick portion **11b** and the vibrator **11a** is easily reduced compared with the other case. A step may be provided instead of the inclined portion.

(18) An angle θ is formed between an inclined surface of the inclined portion **11ba** and the flat surface formed by the vibrator **11a** and the first thick portion **11b**. In this case, the range is $\theta=35\pm 3$ degrees. Since the inclined surface having the angle θ is also a surface derived from a crystal face of quartz-crystal generated in the formation of the quartz-crystal vibrating piece **10** by a photolithography technique and a wet etching technique, the inclined surface having the angle θ contributes to forming the inclined portion with controllability.

(19) The thickness t_2 of the portion other than the inclined portion **11ba** of the first thick portion **11b** is preferably an appropriate thickness in consideration of reduction of a stress on the vibrator **11a**, a better seating of the element unit **11** itself connected and fixed to a container (see FIG. 3), and the like. Relative to the thickness t_1 of the vibrator **11a**, the thickness t_2 of the first thick portion **11b** is preferably a thickness, for example, meeting $t_2 \geq 10 \cdot t_1$, and more preferably a thickness meeting $t_2 \geq 15 \cdot t_1$, although not limited thereto. The upper limit of the thickness t_2 is determined depending on an internal height of a container in which the quartz-crystal vibrating piece **10** is mounted, an economical thickness of the crystal wafer used for the manufacture of the quartz-crystal vibrating piece **10**, and the like.

(20) It is preferred that a dimension δ along a first direction **D1** of the inclined portion **11ba** is long to some extent because the vibrator **11a** can be spaced from the first thick portion **11b**. For example, $\delta > t_2$ is preferred. However, the upper limit of the dimension δ is determined in consideration of the whole dimension in the first direction **D1** of the quartz-crystal vibrating piece **10**, the thickness of the crystal wafer to form the quartz-crystal vibrating piece **10**, the required oscillation frequency, and the like.

(21) The second thick portions **11c** are connected both side surfaces of an end portion of the first thick portion **11b** in the opposite side of the vibrator **11a**, and each extend along a second direction **D2** intersecting with the first direction **D1**. Moreover, the second thick portions **11c** have a thickness t_3 thicker than the thickness t_1 of the vibrator **11a** and same as or different from the thickness t_2 of the first thick portion, and the second thick portions **11c** are used for connecting between the quartz-crystal vibrating piece **10** and, for example, a container (see FIG. 3 and the like) as an external member.

(22) Here, a direction in which the first direction **D1** intersects with the second direction **D2** is configured to be, in this embodiment, a direction in which the first direction **D1** is perpendicular to the second direction **D2**. When the first direction **D1** and the second direction **D2** are perpendicular to one another, and the quartz-crystal vibrating piece **10** is an AT-cut crystal element, the first direction **D1** is preferably a direction parallel to an X-axis of a crystallographic axis of quartz-crystal, and the second direction **D2** is preferably is a direction parallel to a Z'-axis of the crystallographic axis of the quartz-crystal. That is, a direction in which what is called an X-long quartz-crystal vibrating piece is obtained as the quartz-crystal vibrating piece **10** is preferred. On the contrary, the first direction **D1** may be a direction parallel to the Z'-axis and the second direction **D2** may be a direction parallel to the X-axis. That is, it may be a direction in which what

is called a Z-long quartz-crystal vibrating piece is obtained as the quartz-crystal vibrating piece **10**. The first direction **D1** and the second direction **D2** may intersect with one another at a non-perpendicular angle.

(23) The second thick portion **11c** of the embodiment is connected to the first thick portion **11b** with a dimension γ along the first direction **D1**, extends in the second direction with a dimension L , and has a quadrilateral shape in plan view. Here, the dimension L preferably meets $L \geq 0.5 W$ relative to a width W in the second direction of a portion in contact with the second thick portion **11c** of the first thick portion **11b** (in this embodiment, a dimension same as the length α of one side of the vibrator **11a**). This is because the longer the dimension L is, the more the portion connected to the external member of the second thick portion **11c** can be spaced from the vibrator **11a**. The upper limit of the dimension L is preferably determined, for example, in consideration of a limitation of the dimension in the second direction **D2** of the quartz-crystal vibrating piece **10**. While the lengths L of the second thick portions **11c** disposed in the right and the left of the first thick portion **11b** are same in this embodiment, they may be different from one another.

(24) The dimension γ preferably meets $\gamma < \beta$ relative to the dimension β of the first thick portion **11b**. Meeting $\gamma < \beta$ is preferred because the connection width γ of the second thick portion **11c** to the first thick portion **11b** can be specified relative to the dimension β with a predetermined relation. However, when the dimension γ is small, the strength of the second thick portion **11c** as a securing portion cannot be ensured, and therefore, the lower limit of the dimension γ is equal to or more than a dimension with which the above-described strength can be ensured.

(25) The second thick portions **11c** have the thickness $t3$ thicker than the thickness $t1$ of the vibrator **11a** and same as or different from the thickness $t2$ of the first thick portion **11b**. When the thickness of the second thick portion **11c** is the thickness $t3$, the thickness $t3$ may be thicker or thinner than the thickness $t2$ of the first thick portion **11b** on the premise that the thickness $t3$ is thicker than the thickness $t1$ of the vibrator **11a**. However, in this embodiment, the thickness $t1$ of the vibrator **11a**, the thickness $t2$ of the first thick portion **11b**, and the thickness $t3$ of the second thick portion **11c** meet $t1 < t2 = t3$.

(26) The excitation electrodes **13** are disposed on both front and back sides of the vibrator **11a**. A planar shape of the excitation electrode **13** may be any given preferred shape corresponding to the design, for example, a quadrilateral shape, a circular shape, and an elliptical shape, and in this embodiment, the excitation electrode **13** has a circular shape in plan view having a diameter ϕ . The excitation electrode **13** has a plane area smaller than the vibrator **11a** having the square shape in plan view ($\phi < \alpha$), and disposed on both front and back sides of the vibrator **11a** in a manner in which the center points are matched to one another, the size is same between the front and back sides, and the front and back sides are mutually opposed. However, the excitation electrode **13** may be disposed such that the center point of the excitation electrode **13** is displaced from the center point of the vibrator **11a** by a predetermined dimension in what is called an eccentric state. The excitation electrode **13** can be configured of any given preferred metal film.

(27) The extraction electrode **15** is disposed from the excitation electrode **13** to the second thick portion **11c** via the first thick portion **11b**. Moreover, the extraction electrode **15** is disposed in an end region in the short side direction of the first thick portion **11b** along the first direction **D1**. While the extraction electrodes **15** are disposed on both front and back sides of the vibrator **11a** to extend from the excitation electrodes **13** to the respective second thick portions **11c**, the extraction electrode disposed on the flat surface in which the vibrator **11a** is continuous with the first thick portion **11b** is extended to an opposite surface of the second thick portion **11c** via a side surface of the second thick portion **11c**. This is for convenience in connecting and fixing an electrode included in a container to configure a crystal unit (details will be described below) to the extraction electrode **15** with a conductive adhesive or the like. The extraction electrode **15** can be formed of any given preferred metal film, and typically, integrally formed with the excitation electrode **13**.

2. Working Example of Quartz-Crystal Vibrating Piece

(28) As a working example of the quartz-crystal vibrating piece **10**, a quartz-crystal vibrating piece having dimensions of main parts as exemplified below was experimentally produced.

(29) The dimension α of the one side of the vibrator **11a** was 0.4 mm, the dimension β of the first thick portion **11b** other than the portion in contact with the second thick portion **11c** was 0.7 mm, the dimension γ of the second thick portion **11c** in contact with the first thick portion **11b** was 0.3 mm, the dimension L in the second direction D2 of the second thick portion **11c** was 0.2 mm, the dimension W along the second direction D2 of the portion in contact with the second thick portion **11c** of the first thick portion **11b** was 0.4 mm (that is, the same as the dimension α), the dimension δ of the inclined portion was 0.07 mm, the thickness t_1 of the vibrator **11a** was a thickness corresponding to the oscillation frequency of 460 MHz (fundamental wave), the thickness t_2 of the first thick portion **11b** and the thickness t_3 of the second thick portion **11c** were each 0.05 mm, the diameter ϕ of the excitation electrode **13** was 0.1 mm, and the angle θ was about 35 degrees. A large number of the above-described quartz-crystal vibrating pieces were experimentally produced.

(30) The above-described dimension values are target dimensions. In the examination with the target dimensions, $\beta > \alpha$ is met with $\beta/\alpha = 0.7/0.4 = 1.75$. $\gamma < \beta$ is met with $\gamma/\beta = 0.3/0.7 = 0.43$. $L = W$ is met. $\delta > t_2$ is met with $\delta/t_2 = 0.07/0.05 = 1.4$. Since the dimension δ is 0.07 mm, and the thickness t_2 is 0.05 mm, the angle θ meets $0.05/0.07 \approx 0.714 \approx 35.5$ degrees.

(31) The experimentally produced quartz-crystal vibrating piece is hereinafter referred to as an experimentally produced quartz-crystal vibrating piece **10x**. In a plurality of the experimentally produced quartz-crystal vibrating pieces **10x**, each of the dimensions $\alpha, \beta, \gamma, \delta$, and L was a dimension in a range of 1% as a predetermined tolerance.

(32) A crystal unit was produced using the experimentally produced quartz-crystal vibrating piece **10x**, and an oscillation frequency change over time and a drive current change over time of the crystal unit were examined. This will be described in detail in the next section of embodiment of crystal unit.

(33) An analytical model for a finite element method was produced based on the shape dimensions of the experimentally produced quartz-crystal vibrating piece **10x**, and a stress distribution was analyzed in a structure in which this model was fixed at a portion corresponding to the second thick portion in a ceramic package with a polyimide-based conductive adhesive. The analysis was performed by examining a stress distribution generated in the analytical model when the temperature of the analytical model was changed from room temperature (25° C.) to 125° C.

(34) FIG. 2 is a drawing illustrating the analyzed stress distribution. In FIG. 2, the darker gray is, the larger the influence of the stress at the adhesive portion. As specific values, a von Mises stress at a center Q1 in a region in contact with the second thick portions **11c** of the first thick portion **11b** is 4.8×10^7 (10 to the 7th power). The von Mises stress at a center Q2 in a region in the second thick portions **11c** side from a center line P of the first thick portion **11b** excluding the region in contact with the second thick portions **11c** is 3.5×10^6 . The von Mises stress at a center Q3 in a region in the vibrator **11a** side from the center line P of the first thick portion **11b** excluding the region in contact with the second thick portions **11c** is 2.8×10^5 . The von Mises stress at a center Q4 of the vibrator **11a** is 2.6×10^4 . It is seen that the stress at the vibrator **11a** is smaller than the stress in the region in contact with the second thick portions **11c** of the first thick portion **11b** by three orders of magnitude. Moreover, it is seen that in the first thick portion **11b**, the influence of the stress decreases toward the vibrator **11a** from near the center P (see FIG. 2) in the longitudinal direction of the first thick portion **11b**. From these points, it is seen that the stress in the portion of the vibrator **11a** is small compared with the other portions. Since it can be said that the stress from the supporting portion decreases at around the center in the first direction D1 of the first thick portion **11b**, that is, about $\beta/2 \approx \alpha$, it is seen also from this point that meeting $\beta > \alpha$ is preferred.

3. Embodiment of Aspect of Crystal Unit

(35) Next, an embodiment of a crystal unit as a second aspect of this application will be described.

FIG. 3 is a perspective view for describing a crystal unit **30** of the embodiment, and is a perspective view illustrated by partially cutting away a container **31** included in the crystal unit **30**.

(36) The crystal unit **30** of the embodiment includes the quartz-crystal vibrating piece **10** of the first aspect, the container **31** that houses the quartz-crystal vibrating piece **10**, and a lid member **33** that seals the container **31**. The quartz-crystal vibrating piece **10** is electrically connected and fixed to the container **31** at the second thick portions **11c**.

(37) In this embodiment, the container **31** is provided with a depressed portion **31a** in which the quartz-crystal vibrating piece **10** is housed, a wall portion **31b** surrounding the depressed portion **31a**, adhesion pads **31c** disposed at positions corresponding to the second thick portions **11c** of the quartz-crystal vibrating piece **10** on a bottom surface of the depressed portion **31a**, and external connecting terminals **31d** externally disposed on the bottom surface of the container **31**. The adhesion pads **31c** are connected to the external connecting terminals **31d** by via-wirings or the like (not illustrated). The container **31** can be configured of, for example, a ceramic package.

(38) The quartz-crystal vibrating piece **10** is connected and fixed to the adhesion pads **31c** of the container **31** with a conductive adhesive **35** at the positions of the extraction electrodes **15** of the second thick portions **11c**. Then, the crystal unit **30** of the embodiment is configured by sealing an opening of the depressed portion **31a** with the lid member **33**. The structure and the shape of the lid member **33** can be chosen corresponding to the sealing method. In this embodiment, the lid member **33** is configured of a flat plate-shaped member.

(39) A plurality of crystal units were experimentally produced using the experimentally produced quartz-crystal vibrating piece **10x** as the quartz-crystal vibrating piece **10** embedded into the crystal unit **30** described with reference to FIG. 3. Then, the experimentally produced crystal units were left to be non-operated in an oven set to the temperature at 125° C., and extracted from the oven at approximate timings, and changes over time of the frequency and the drive current were each measured.

(40) FIG. 4A is a drawing illustrating the frequency change over time in the above-described test, and FIG. 4B is a drawing illustrating the drive current change over time in the above-described test. In both drawings, the horizontal axis indicates the elapsed time, and the vertical axis indicates the change rate of data relative to an initial value in each measurement, that is, a frequency change rate (ppm) or a current change rate (%). From both drawings, it is seen that at a time point at which about 2500 hours has elapsed after the start of the test, the frequency change rate is ± 5 ppm and the change rate tends to decrease, and the drive current change rate is about 1% and tends to saturate. Therefore, it is seen that the experimentally produced crystal unit exhibits practical characteristics.

(41) While a container with a depressed portion is used as the container **31**, and a flat plate-shaped one is used as the lid member **33** in the above-described configuration, a flat plate-shaped one may be used as the container, and a cap-shaped lid member with a depressed portion to house the quartz-crystal vibrating piece may be used as the lid member.

4. Embodiment of Aspect of Crystal Controlled Oscillator

(42) Next, an embodiment of a crystal controlled oscillator as a third aspect of this application will be described. FIG. 5 is a perspective view for describing a crystal controlled oscillator **40** of the embodiment, and is a perspective view illustrated by partially cutting away a container **43** included in the crystal controlled oscillator **40**.

(43) The crystal controlled oscillator **40** of the embodiment includes the quartz-crystal vibrating piece **10** of the first aspect, an oscillator circuit **41** for the quartz-crystal vibrating piece, the container **43** that houses the quartz-crystal vibrating piece **10** and the oscillator circuit **41**, and the lid member **33** that seals the container **43**. The quartz-crystal vibrating piece **10** is electrically connected and fixed to the container **43** at the second thick portions **11c**.

(44) In this embodiment, the container **43** is provided with a second depressed portion **43a** to house the oscillator circuit **41** at the bottom surface of the depressed portion **31a** that houses the quartz-crystal vibrating piece **10**. The other structure is similar to the structure of the above-described

container **31**.

(45) The oscillator circuit **41** may be any given preferred one, and may be configured of an integrated circuit having an oscillating function and a temperature compensation function, or the like.

(46) In the above-described example, a crystal controlled oscillator of what is called a one-chamber structure, in which the quartz-crystal vibrating piece **10** and the integrated circuit including the oscillator circuit are mounted in one container, is described. However, although not illustrated, the present disclosure is applicable to a crystal controlled oscillator of what is called an H-type structure in which a crystal chamber that houses the quartz-crystal vibrating piece and an IC chamber that houses a semiconductor integrated circuit for an oscillator circuit are stacked.

5. Intermediate Wafer for Forming Crystal Resonator

(47) Next, an embodiment of an intermediate wafer for forming a crystal resonator as a fourth aspect of this application will be described.

(48) FIG. **6** is a top view for describing an intermediate wafer **50** for forming a crystal resonator (hereinafter abbreviated as intermediate wafer **50**) of the embodiment and its partially enlarged view. FIG. **6** illustrates a crystal wafer as the intermediate wafer **50** with a relation between crystallographic axes of quartz-crystal of an X-axis, Y'-axis, and Z'-axis.

(49) The intermediate wafer **50** of the embodiment is configured of a crystal wafer **50** including a large number of the quartz-crystal vibrating pieces **10** of the first aspect in a matrix. While the intermediate wafer **50** has a circular planar shape in the drawing, the planar shape may be a quadrilateral shape. In the intermediate wafer **50**, each of the quartz-crystal vibrating pieces **10** is connected to a frame portion **51** via a snap-off portion **53**. The respective quartz-crystal vibrating pieces **10** are snapped off from the intermediate wafer **50** at the positions of the snap-off portions **53**, diced to the quartz-crystal vibrating pieces **10**, and further, can be used by being mounted to the containers of the crystal unit and the crystal controlled oscillator.

6. Modifications of Quartz-Crystal Vibrating Piece

(50) The quartz-crystal vibrating piece of the first aspect may be modified, for example, as described below. FIG. **7A** to FIG. **7E** are explanatory drawings thereof, FIG. **7A** to FIG. **7D** are top views of crystal units in the respective modifications, and FIG. **7E** is a side view of a crystal unit of a modification.

(51) As illustrated in FIG. **7A**, a quartz-crystal vibrating piece **71** of Modification 1 is provided with cutouts from the right and the left toward the center along a short side direction at an appropriate position in the longitudinal direction of the first thick portion **11b**, and a portion between the cutouts is formed as a narrowed portion **71a**. While one narrowed portion **71a** is provided in the drawing, a plurality of the narrowed portions **71a** may be provided.

(52) As illustrated in FIG. **7B**, a quartz-crystal vibrating piece **72** of Modification 2 is provided with a through hole **72a** at an appropriate position in the longitudinal direction of the first thick portion **11b**. While one through hole **72a** is provided in the drawing, a plurality of the through holes **72a** may be provided.

(53) As illustrated in FIG. **7C**, a quartz-crystal vibrating piece **73** of Modification 3 is provided with cutouts from upper and lower sides toward the center along the longitudinal direction of the first thick portion **11b** at each of the second thick portions **11c**, and a portion between the cutouts is formed as a narrowed portion **73a**. A plurality of numbers of the narrowed portions **73a** may be provided.

(54) As illustrated in FIG. **7D**, a quartz-crystal vibrating piece **74** of Modification 4 is provided with a through hole **74a** at an appropriate position of the second thick portion **11c**. A plurality of numbers of the through holes **74a** may be provided.

(55) In each of Modification 1 to Modification 4, since the narrowed portion or the through hole is provided between the vibrator **11a** and the connection point with the external member of the second thick portion **11c**, the mutual interference with the connection point with the external member of

the second thick portion **11c** is considered to be easily reduced.

(56) A quartz-crystal vibrating piece **75** of Modification 5 is an example of a structure in which inclined portions **75a**, **75b** or step portions (not illustrated) are provided from both sides in the direction of the thickness **t2** in the vibrator side of the first thick portion **11b**, and the vibrator **11a** is connected to the first thick portion **11b** halfway in the thickness **t2** of the first thick portion **11b**. The connecting position of the vibrator **11a** to the first thick portion **11b** may be the middle in the direction along the thickness **t2**, and may be a position close to one side. In the case of Modification 5, since the vibrator **11a** is connected to the first thick portion in a state where the inclined portions or the like are provided in both sides in the direction of the thickness **t2**, it is considered that edge cutting of the vibrator from the first thick portion is easily performed.

(57) According to a second aspect of this disclosure, there is provided a crystal unit that includes the quartz-crystal vibrating piece according to the first aspect, and a container as the external member. The container houses the quartz-crystal vibrating piece. The quartz-crystal vibrating piece is electrically connected and fixed to the container at the second thick portions.

(58) According to a third aspect of this disclosure, there is provided a crystal controlled oscillator that includes the quartz-crystal vibrating piece according to the first aspect, an oscillator circuit for the quartz-crystal vibrating piece, and a container as the external member. The container houses the quartz-crystal vibrating piece and the oscillator circuit. The quartz-crystal vibrating piece is electrically connected and fixed to the container at the second thick portions.

(59) According to a fourth aspect of this disclosure, there is provided an intermediate wafer for forming a quartz-crystal vibrating piece. The intermediate wafer includes a crystal wafer including a large number of the quartz-crystal vibrating pieces according to the first aspect in a matrix.

(60) To embody the first to the fourth aspects, in the quartz-crystal vibrating piece, when a dimension along the first direction of the vibrator is α , a dimension along the first direction of a portion of the first thick portion excluding a portion in contact with the second thick portions is β , a dimension along the first direction of a portion of the second thick portion in contact with the first thick portions is γ , a dimension along the second direction of a portion of the first thick portion in contact with the second thick portions is W , and a dimension along the second direction of the second thick portion is L , although details will be described in the embodiments below, there are preferably predetermined relations between a part of or all of these dimensions α , β , γ , W , and L . This allows reducing the mutual interference between the vibrator and the external member compared with the other case.

(61) To embody the first to the fourth aspects, in the quartz-crystal vibrating piece, the first thick portion preferably includes an inclined portion having a thickness gradually thinned toward the vibrator in the vibrator side. When the inclined portion is provided, the mutual interference between the vibrator and the external member can be reduced compared with the other case.

(62) The quartz-crystal vibrating piece of this disclosure can achieve the quartz-crystal vibrating piece having the structure in which the vibrator is spaced from the supporting portion of the container or the like by the first thick portion and the second thick portion intersecting with the first thick portion. Moreover, the quartz-crystal vibrating piece is connected to the outside via the second thick portion. From these points, the influence of the external stress on the vibrator can be reduced compared with conventional one. Accordingly, the quartz-crystal vibrating piece, the crystal unit, and the crystal controlled oscillator having the novel structure appropriate for achieving a high frequency quartz crystal device can be provided.

(63) The principles, preferred embodiment and mode of operation of the present invention have been described in the foregoing specification. However, the invention which is intended to be protected is not to be construed as limited to the particular embodiments disclosed. Further, the embodiments described herein are to be regarded as illustrative rather than restrictive. Variations and changes may be made by others, and equivalents employed, without departing from the spirit of the present invention. Accordingly, it is expressly intended that all such variations, changes and

equivalents which fall within the spirit and scope of the present invention as defined in the claims, be embraced thereby.

Claims

1. A quartz-crystal vibrating piece comprising: an element unit including: a vibrator that has a quadrilateral shape in plan view, vibrates in a thickness-shear mode, and has a thickness t_1 ; a first thick portion that has a rectangular shape in plan view, has a short side connected to one side of the vibrator, extends in a first direction, and has a thickness t_2 thicker than the thickness t_1 ; and second thick portions connected to both side surfaces of an end portion of the first thick portion in an opposite side of the vibrator, each of the second thick portions extending along a second direction intersecting with the first direction and having a thickness t_3 thicker than the thickness t_1 and same as or different from the thickness t_2 , the second thick portions being used for connecting the quartz-crystal vibrating piece to an external member; excitation electrodes disposed on both front and back sides of the vibrator; and an extraction electrode provided from the excitation electrode to the second thick portion via the first thick portion, wherein, when a dimension along the first direction of the vibrator is a , and a dimension along the first direction of a portion of the first thick portion excluding a portion in contact with the second thick portions is β , $\beta > \alpha$ is met.
2. The quartz-crystal vibrating piece according to claim 1, wherein when a dimension along the first direction of a portion of the second thick portion in contact with the first thick portions is γ , $\gamma < \beta$ is met.
3. The quartz-crystal vibrating piece according to claim 1, wherein when a dimension along the second direction of a portion of the first thick portion in contact with the second thick portions is W , and a dimension along the second direction of the second thick portion is L , $L \geq 0.5 W$ is met.
4. The quartz-crystal vibrating piece according to claim 1, wherein when a dimension along the first direction of a portion of the second thick portion in contact with the first thick portions is γ , $\gamma < \beta$ is met, and when a dimension along the second direction of a portion of the first thick portion in contact with the second thick portions is W , and a dimension along the second direction of the second thick portion is L , $L \geq 0.5 W$ is met.
5. The quartz-crystal vibrating piece according to claim 1, wherein when a dimension along the first direction of a portion of the second thick portion in contact with the first thick portions is γ , $\gamma < \beta$ is met, when a dimension along the second direction of a portion of the first thick portion in contact with the second thick portions is W , and a dimension along the second direction of the second thick portion is L , $L \geq 0.5 W$ is met, and the thickness t_2 meets $t_2 \geq 10 \cdot t_1$ relative to the thickness t_1 .
6. The quartz-crystal vibrating piece according to claim 1, wherein the first thick portion includes an inclined portion having a thickness gradually thinned toward the vibrator in the vibrator side.
7. The quartz-crystal vibrating piece according to claim 1, wherein the first thick portion includes an inclined portion having a thickness gradually thinned toward the vibrator in the vibrator side, and when a dimension along the first direction of the inclined portion is defined as δ , the dimension δ meets $\delta > t_2$ relative to the thickness t_2 .
8. The quartz-crystal vibrating piece according to claim 1, wherein the first thick portion includes an inclined portion having a thickness gradually thinned toward the vibrator in the vibrator side, one surface of the vibrator and one surface of the first thick portion constitute a continuous flat surface, and an angle θ formed between an inclined surface of the inclined portion and the flat surface is $\theta = 35 \pm 3$ degrees.
9. The quartz-crystal vibrating piece according to claim 1, wherein when a dimension along the first direction of a portion of the second thick portion in contact with the first thick portions is γ , $\gamma \leq \beta$ is met, when a dimension along the second direction of a portion of the first thick portion in contact with the second thick portions is W , and a dimension along the second direction of the

second thick portion is L , $L \geq 0.5 W$ is met, the thickness t_2 meets $t_2 \geq 10 \cdot t_1$ relative to the thickness t_1 , and the first thick portion includes an inclined portion having a thickness gradually thinned toward the vibrator in the vibrator side.

10. A crystal unit comprising: the quartz-crystal vibrating piece according to claim 1; and a container as the external member, the container housing the quartz-crystal vibrating piece, and the quartz-crystal vibrating piece being electrically connected and fixed to the container at the second thick portions.

11. A crystal controlled oscillator comprising: the quartz-crystal vibrating piece according to claim 1; an oscillator circuit for the quartz-crystal vibrating piece; and a container as the external member, the container housing the quartz-crystal vibrating piece and the oscillator circuit, the quartz-crystal vibrating piece being electrically connected and fixed to the container at the second thick portions.

12. An intermediate wafer for forming a quartz-crystal vibrating piece, comprising a crystal wafer including a large number of the quartz-crystal vibrating pieces according to claim 1 in a matrix.
